BEN CLAYDON, MECHANICAL ENGINEERING STUDENT

Portfolio Website: ben.claydon.ca **Phone**: (403) 519-8525 **Email**: ben@claydon.ca

EDUCATION

Bachelor of Engineering in Mechanical Engineering (BEng) **Sep 2022 – Aug 2027 (Expected)**
University of Victoria, Victoria, BC

COMPETENCIES

- **Problem Solving** – Determined the braking force of an undocumented brake function in autonomous vehicle software using timed testing and kinematic equations.
- **Attention to Detail** – As an intern at EnviroSearch, identified previously unnoticed calculation errors in spreadsheet data and proposed a solution to fix them.
- **Natural Leader** – Frequently perform leadership roles and project management duties in group and club work by setting up communication channels, delegating tasks, and collaborating with members to help them complete their tasks independently.
- **Technical Proficiency** – SolidWorks; Mastercam; MATLAB; Bluebeam; Linux (Ubuntu) OS; ROS 2; Docker; Coding in C, C++, Python; VBA; MS Excel; MS Word; Adobe Acrobat; AutoCAD; Claude AI

WORK EXPERIENCE

Co-op Student **Jun 2025 – Dec 2025**
Sequoia Environmental Remediation Inc., Calgary, AB

- Performed hands-on assistance with manufacturing and maintenance of environmental remediation equipment involving mechanical and electrical systems.
- Redesigned a pump control system with a voltage boost converter, extending the pump's maximum operating depth.
- Designed and 3D printed a detailed model of a chemical injection system for marketing conferences, a handheld mount for a GPS transmitter and antenna, and a mounting/shielding device for an infrared transmitter.
- Created SolidWorks models and engineering drawings for a custom air purification system tailored to client needs. Learned rendering after hours to produce photorealistic images of the design.
- Performed design updates remotely after the co-op end date to accommodate unanticipated design changes.

Co-op Student **May 2024 – Aug 2024**
RJC Engineers, Victoria, BC

- Wrote a VBA script (with an instruction manual) that automatically formats 10-page meeting-minute reports, reducing editing time by 30%.
- Created a three-stage process roadmap for funding program applications on behalf of RJC clients.
- Implemented the use of a DJI Mini 4 Pro drone to capture aerial images for roof assessments, saving the time and cost of accessing roofs to manually take photos.
- Created a 3D interactive photo map of the B.C. Parliament Buildings.
- Used advanced Excel formulas to automate spreadsheet generation in large 40-sheet workbooks.
- Performed on-site assessments of roof structures and building enclosures.
- Conducted airtightness testing and water penetration testing on buildings.
- Communicated with clients and contractors in construction meetings to manage progress of ongoing construction projects.

Summer Student/Junior Technician **May 2023 – Sep 2023**
EnviroSearch Ltd., Calgary, AB

- Operated peristaltic pumps, inertial pumps, and a multiparameter water quality meter to collect groundwater samples and measure field parameters at contaminated sites across rural Alberta, following standard operating procedures.
- Developed and facilitated a well-repair project across 20 sites.
- Proposed and implemented a new method of calcium peroxide cell deployment to improve remediation of wells with changing water tables.
- Created technical reports, groundwater and soil tables, work summaries, relative percent difference tables, and photo logs in Microsoft Word and Excel.

DESIGN TEAM – CONNECTED AND AUTONOMOUS VEHICLE CLUB

Club Vice President & Team Leader **Sep 2024 – Present**

- Manage a design team of graduate and undergraduate students that converts remote-controlled 1/10 scale cars to autonomous driving vehicles and competes in racing competitions.
- Initiated a club restructure to improve technical progress and member engagement.
- Developed a reactive algorithm that continuously computes and publishes the desired steering angle and speed of the car based on lidar input.
- Developing a SLAM (Simultaneous Localization and Mapping) algorithm that generates a map of the course based on lidar data and computes the optimal racing line for the car to follow.

Team Member **Jun 2024 – Aug 2024**

- Promoted to Club Vice President and Team Leader for contributions to team success.

DESIGN PROJECTS

- Photos, drawings, and write-ups for each project are on my portfolio website: ben.claydon.ca
- Some projects include: Automated Part Sorting System, 1/10 Scale Autonomous Race Car, Custom Air Purification System and Chemical Injection Skid Model.

INTERESTS, HOBBIES

- Outdoor Activities – Freestyle Skiing, Camping, Hiking, Surfing, Climbing
- Music – Played guitar and bass guitar in live shows as a member of a band.
- Sports – Founder of annual squash tournament, captain of two intramural volleyball teams, member of intramural soccer and ultimate frisbee teams.

AWARDS

- IEEE F1Tenth Autonomous Racing Challenge (Victoria Section) – Champion Award, Summer 2024; Technical Excellence Award, Summer and Fall 2024; Best Team Collaboration Award, Fall 2024
- Certificate of Achievement, Introduction to Probability and Statistics I, 2024, Dept. of Math & Stats
- Certificate of Achievement, Calculus I, 2022, Dept. of Math & Stats
- University of Victoria Entrance Scholarship (John and Myrtle Tilley Undergraduate Scholarship), 2022

CERTIFICATIONS

- Alberta Class 5 Driver's Licence
- Energy Safety Canada Common Safety Orientation
- WHMIS (GHS) (expires 2028-06-02)
- TDG Training (expires 2028-06-02)

REFERENCES

Kevin Keenan, P. Geol.
Vice President, Sequoia Environmental Remediation Inc.
(403) 862-4309, kevin@sequoiaenvironmental.com

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